

Pakistan Stock Exchange (PSX) — Snowflake + dbt + Power BI

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Course: Data Warehousing & Business Intelligence

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Project: End-to-End Data Warehouse using Snowflake, dbt, and Power BI

Dataset Size: 49,820 rows

Tools Used: Snowflake, dbt, VS Code, Power BI



1. Introduction

This project demonstrates the complete lifecycle of building a modern cloud-based **Data Warehouse** using:

- **Snowflake** (Cloud Data Platform)
- **dbt** (Data transformation & automation)
- **Power BI** (Business Intelligence & Analytics)

The objective is to convert raw Pakistan Stock Exchange (PSX) data into structured dimensional models (facts & dimensions), automate transformations, test data quality, and build final dashboards for financial insights.

2. Project Objectives

The main goals of the project were:

1. Load and manage PSX stock price data in Snowflake
 2. Build a scalable multi-layer Data Warehouse
 3. Use dbt for transformations and documentation
 4. Create analytic models (fact & dimension tables)
 5. Perform testing and validation
 6. Build dashboard insights in Power BI
 7. Present final business insights to decision makers
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3. Dataset Description

The dataset contains **49,820 rows** with daily stock information from 2021 to 2024. Columns include:

- **Date**
- **Symbol**
- **Open Price**
- **High Price**
- **Low Price**

- **Close Price**

The dataset was mostly clean and required only structural standardization, renaming, and date formatting.

4. System Architecture

The architecture follows industry-standard layers:

1. RAW Layer

- Loaded directly from CSV
- Stored exactly as provided (no changes)

2. STAGING Layer

- Cleans data
- Standardizes date format
- Fixes column names
- Removes duplicates

3. DIMENSION Layer

- dim_stock (symbol reference)
- dim_date (calendar table)

4. FACT Layer

- fact_stock_prices
- fact_market_summary

5. Power BI Reporting Layer

- Dashboards and visual insights
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5. Snowflake Setup

Database & Schemas Created

PSX_DB
└─ RAW
└─ STAGING
└─ DWH

Warehouse

PROJECT_WH (Small)

Loading Dataset

The file *psx_data_20211210_to_20241210.csv* was uploaded into:

PSX_DB.RAW.PSX_DATA

Total rows loaded: **49,820**

6. dbt Project Setup (VS Code)

Steps:

1. Installed dbt-core & dbt-snowflake
2. Ran: dbt init psx_project
3. Configured profiles.yml for Snowflake login
4. Verified connection with dbt debug
5. Created folder structure:

```
models/  
├─ sources/  
├─ staging/  
├─ dim/  
├─ facts/  
└─ tests/
```

7. dbt Models Implementation

7.1 RAW Source Definition

File: psx_sources.yml

sources:

- name: psx_raw
database: PSX_DB
schema: RAW
tables:
- name: PSX_DATA

7.2 STAGING Model

File: stg_psx_data.sql

```
SELECT DISTINCT
  TO_DATE(Date) AS trade_date,
  Symbol AS stock_symbol,
  Open AS open_price,
  High AS high_price,
  Low AS low_price,
  Close AS close_price
FROM {{ source('psx_raw', 'PSX_DATA') }};
```

Purpose:

- Standardize naming
 - Clean data
 - Convert date format
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7.3 DIMENSION MODELS

dim_stock

```
SELECT DISTINCT
  stock_symbol,
  INITCAP(stock_symbol) AS stock_name
FROM {{ ref('stg_psx_data') }};
```

dim_date

A full calendar table used for time-based reporting.

```
WITH dates AS (
  SELECT DATEADD(day, seq4(), '2010-01-01') AS cal_date
  FROM TABLE(GENERATOR(ROWCOUNT => 6000))
)
SELECT
  cal_date AS date_key,
```

```
YEAR(cal_date) AS year,  
MONTH(cal_date) AS month,  
DAY(cal_date) AS day,  
MONTHNAME(cal_date) AS month_name,  
DAYOFWEEK(cal_date) AS day_of_week,  
WEEKOFYEAR(cal_date) AS week_no  
FROM dates;
```

7.4 FACT MODELS

fact_stock_prices

```
SELECT  
  s.stock_symbol,  
  d.date_key,  
  t.open_price,  
  t.high_price,  
  t.low_price,  
  t.close_price  
FROM {{ ref('stg_psx_data') }} t  
JOIN {{ ref('dim_stock') }} s  
  ON t.stock_symbol = s.stock_symbol  
JOIN {{ ref('dim_date') }} d  
  ON t.trade_date = d.date_key;
```

fact_market_summary

```
SELECT  
  stock_symbol,  
  YEAR(trade_date) AS year,  
  MONTH(trade_date) AS month,  
  AVG(close_price) AS avg_close,  
  MAX(close_price) AS max_close,  
  MIN(close_price) AS min_close,  
  SUM(high_price - low_price) AS volatility  
FROM {{ ref('stg_psx_data') }}  
GROUP BY 1,2,3;
```

8. Data Quality Testing (dbt tests)

Implemented tests:

Uniqueness Tests

- stock_symbol

- date_key

Not Null Tests

- close_price
- trade_date

Relationship Tests

- fact tables must match dimension tables

Command used:

```
dbt test
```

All tests passed successfully.

9. Lineage & Documentation

Used:

```
dbt docs generate  
dbt docs serve
```

Generated:

- Complete lineage graph
- Auto-documented models
- Column-level metadata
- Full transformation flow

This provides professional-grade project transparency.

10. Power BI Dashboard Development

Connected Power BI to Snowflake using:

- Server: account.region.snowflakecomputing.com
- Warehouse: PROJECT_WH

- Database: PSX_DB
- Schema: DWH

Dashboards Created

1. **Daily Stock Price Trends**
2. **Yearly Market Performance**
3. **Volatility Analysis**
4. **Stock Comparison Chart**
5. **Monthly Aggregated Summary**

Business users can now analyze:

- Trend patterns
 - Volatility spikes
 - Best & worst performing stocks
 - Seasonal variations
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11. Key Insights

Some highlights discovered:

- Certain stocks show strong upward trends during specific months
 - Volatility increases around economic events
 - Closing prices show periodic stable patterns
 - Some stocks maintain consistently low volatility
 - Monthly averages help predict stable investment choices
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12. Conclusion

This project successfully demonstrates:

End-to-end cloud data warehouse development
Automated ELT pipeline using dbt
Clean dimensional modeling (FACT + DIM)
High-quality documentation and tests
Professional dashboards using Power BI

The combination of **Snowflake + dbt + Power BI** provides a robust and industry-standard analytics stack suitable for real business use.